Discrete Mathematics — Study Notes

Topics: Partitions · Partial Order Relations · Bounds & Posets · Functions — compiled from lecture slides, written in plain English for catch-up revision

Contents

1. Partitions

Big idea: A partition just means "cutting a set into pieces" so that (a) every element of the original set ends up in exactly one piece, (b) no element is left out, and (c) no two pieces overlap. Think of slicing a pizza — every slice is part of the pizza, every point of the pizza belongs to exactly one slice, and slices don't overlap.

Definition — Partition of a set

Let **A** be a set, and let $\mathscr{A} = \{A_n : n \in I\}$ be a collection of *non-empty* subsets $A_n \subseteq A$. We say **A is partitioned by $\mathscr{A}$** if:

1. $\mathbf{A = \bigcup_{n\in I} A_n}$ — the pieces A_n together make up all of A (nothing is missing).
2. **If $A_m \neq A_n$, then $A_m \cap A_n = \varnothing$** — distinct pieces are *pairwise disjoint* (no overlap).

We call the collection $\mathscr{A}$ **a partition of A**.

Note — equivalent way to state it

$\mathscr{A}$ is a partition of a nonempty set A if and only if:

- $\mathscr{A} \subseteq P(A)$ (each piece is a subset of A, i.e. an element of A's power set),
- $\bigcup\mathscr{A} = A$ (union of all the pieces gives back A),
- $\mathscr{A}$ is pairwise disjoint.

Worked Example

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Define $A_0 = \{1, 2, 5\}$, $A_1 = \{3\}$, $A_2 = \{4, 6, 7\}$.
Then $\mathscr{A} = \{A_0, A_1, A_2\} \subseteq P(A)$. Check:

- $A_0 \cup A_1 \cup A_2 = \{1,2,3,4,5,6,7\} = A$ ✔
- $A_0 \cap A_1 = \varnothing$, $A_0 \cap A_2 = \varnothing$, $A_1 \cap A_2 = \varnothing$ ✔ (pairwise disjoint)

Therefore $\mathscr{A}$ **is** a partition of A. (Picture: three separate "blobs" inside a big circle labelled A, with no overlap and every dot from 1–7 sitting in exactly one blob.)

2. Partitions ↔ Equivalence Relations

Big idea: This is one of the most tested ideas in this topic. Every equivalence relation "chops up" a set into equivalence classes, and those classes form a partition. Conversely, if you're handed a partition, you can build an equivalence relation from it (two elements are related exactly when they're in the same piece). So partitions and equivalence relations are really *two views of the same thing*.

Lemma

Let A be a set and ~ an equivalence relation on A. Then:

$$a \sim b \iff [a] = [b]$$

(Two elements are related exactly when their equivalence classes are the same set.)

Note

Distinct equivalence classes are pairwise disjoint: if $[x] \neq [y]$, then $[x] \cap [y] = \varnothing$.

Theorem

Let A be a set.

1. If ~ is an equivalence relation on A, then **A is partitioned by the equivalence classes of ~**.
2. If $\{A_n\}_{n\in I}$ is a partition of A, then the relation ~ on A defined by
   $$a \sim b \iff \exists n \in I \text{ such that } a \in A_n \text{ and } b \in A_n$$

is an equivalence relation.

How to remember it: "Equivalence relation → classes → partition" and "partition → same-piece relation → equivalence relation." They go in a loop, both directions always work.

3. Partial Order Relations

Big idea: A partial order is a rule for comparing elements of a set — like ≤ for numbers — but it doesn't have to be able to compare *every* pair. Some pairs might just be "not comparable" (like comparing {1,3} and {2} by ⊆ — neither contains the other). That's exactly why it's called *partial*.

Definition

A relation $\mathscr{R}$ is a **partial order relation** if it is simultaneously:

- **reflexive** (a $\mathscr{R}$ a for all a),
- **antisymmetric** (if a $\mathscr{R}$ b and b $\mathscr{R}$ a, then a = b),
- **transitive** (if a $\mathscr{R}$ b and b $\mathscr{R}$ c, then a $\mathscr{R}$ c) — on the set A.

A set on which a partial order relation is defined is called a **poset** (partially ordered set).

Examples

• ≤ is a partial order relation on ℕ, ℤ, and ℝ.
• The relation $\mathscr{R}$ = {(a, b) | a ⊆ b} defined on the power set of A = {1, 2, 3} is a partial order relation.

Remark 1

The word "partial" indicates that *not every pair* of elements in a set is necessarily comparable. (Two elements a, b of a poset A are **comparable** if a $\mathscr{R}$ b or b $\mathscr{R}$ a.)

Example — incomparable elements

In the poset (P(A), ⊆) where A = {1, 2, 3}: {1,3} ⊄ {2} and {2} ⊄ {1,3}, so {1,3} and {2} are **not comparable**. On the other hand, {1} ⊆ {1,3}, so those two **are** comparable.

Note — poset notation

If a relation ≤ on A is reflexive, antisymmetric, and transitive, we call (A, ≤) a **poset**. For a, b ∈ A, we write **a ≺ b** to mean *a ≤ b but a ≠ b* (strict version, like < vs ≤).

Example to keep straight: ≤ and = are partial orders on ℝ. But plain < is **not** a partial order, because it fails reflexivity (a < a is never true). Equality = is technically a partial order on any set, but it's a boring one — an equivalence relation is *not* normally a partial order because equivalence relations don't need antisymmetry, and most partial orders aren't symmetric.

4. The Weak-Trichotomy Law

Big idea: Full trichotomy (like for real numbers) says exactly one of $a < b$, $b < a$, $a = b$ is true, for *any* a, b. In a general poset, some pairs might not be comparable at all — so we only get *at most one* of these three holding, not "exactly one." That's the "weak" version.

Theorem [Weak-Trichotomy Law]

If $\preceq$ is a partial order on A, then for all $a, b \in A$, **at most one** of the following is true:

$a \prec b$, $b \prec a$, or $a = b$.

Proof sketch: Suppose $a \prec b$ (so $a \preceq b$ and $a \neq b$). If also $b \prec a$, transitivity gives $a \prec a$ — a contradiction (since $a \prec a$ would force $a \neq a$). So $a \prec b$ rules out $b \prec a$. Similarly $b \prec a$ rules out $a \prec b$. And if $a = b$, then by the very definition of $\prec$ (which requires $a \neq b$), neither $a \prec b$ nor $b \prec a$ can hold. So the three cases are mutually exclusive. ∎

5. Bounds: Least, Greatest, Minimal, Maximal

Big idea: "Least/greatest" element means it beats (or loses to) *every single* element of the set — a strong, absolute requirement. "Minimal/maximal" is weaker — it just means nothing else in the set is strictly below/above it, but it might still be incomparable to some elements. A poset can easily have several minimal elements but at most one least element.

Setup — uniqueness of least/greatest

If m, m' both satisfy $a \preceq m$ and $a \preceq m'$ for all $a \in A$, then $m \preceq m'$ and $m' \preceq m$, so by antisymmetry $m = m'$. This is why we can say "**the** least element" — it's unique if it exists.

Definition

Let $(A, \preceq)$ be a poset and $m \in A$.

- m is the **least element** of A if **$m \preceq a$ for all $a \in A$**.
- m is the **greatest element** of A if **$a \preceq m$ for all $a \in A$**.

There is no guarantee a poset will have a least or greatest element.

Example

A finite set of real numbers has both a least and greatest element (under $\leq$).

$\mathbb{Z}^+$ has a least element (1) but no greatest element.

Both $\{5n : n \in \mathbb{Z}^-\}$ and $\mathbb{Z}^-$ have a greatest element but no least element.

For $B = \{5n : n \in \mathbb{Z}^+\}$: the least element of B is 5, because $5 \in B$ and $5(1) \leq 5n$ for all $n \in \mathbb{Z}^+$ (using that 1 is the least element of $\mathbb{Z}^+$).

Definition — comparable / incomparable

Let $(A, \preceq)$ be a poset and $a, b \in A$. If $a \preceq b$ or $b \preceq a$, then a and b are **comparable**. Elements are **incomparable** if they are not comparable.

Definition — minimal / maximal element

Let $(A, \preceq)$ be a poset and $m \in A$.

- m is a **minimal element** if no $a \in A$ satisfies $a \prec m$ (nothing is strictly below m).
- m is a **maximal element** if no $a \in A$ satisfies $m \prec a$ (nothing is strictly above m).

Key contrast to remember:

- **Least/greatest** = compares favourably against *every* element (strong; unique if it exists).
- **Minimal/maximal** = nothing beats it locally, but it might be incomparable with other elements (a poset can have multiple minimal or maximal elements).

6. Chains, Compatibility, Linear Order, Well-Ordering

Definition — Chain

A subset C of the poset (A, ≤) is a **chain** with respect to ≤ if *every pair* of elements a, b ∈ C is comparable.

Example

Let $A_k = \{x \in \mathbb{Z} : (x-1)(x-2)\cdots(x-k) = 0\}$ for $k \in \mathbb{Z}^+$. So $A_m = \{1, 2, \ldots, m\}$ and $A_n = \{1, 2, \ldots, n\}$. If m ≤ n, then $A_m \subseteq A_n$ (otherwise $A_n \subseteq A_m$) — any two of these sets are comparable by ⊆, so $\{A_k : k \in \mathbb{Z}^+\}$ is a chain with respect to ⊆.

Definition — Linear order (total order)

(A, ≤) is a **linearly ordered set** and ≤ is a **linear order** if A itself is a chain with respect to ≤ — i.e. *every* pair of elements is comparable, no exceptions.

> Since every subset of ℝ is a chain under ≤ (any two real numbers are comparable), ≤ is a linear order on ℝ. Contrast this with (P(A), ⊆), where some subsets are incomparable — so ⊆ is only a *partial*, not linear, order.

Definition — Compatible / Incompatible

Let (A, ≤) be a poset. Elements a, b ∈ A are **compatible** if there exists c ∈ A such that c ≤ a and c ≤ b (they have a common lower bound). If not, they are **incompatible**, written $a \perp b$.

Example 1 — Divisibility poset

Let A = {1, 2, 3, 4, 6, 12}, ordered by divisibility: $a \preceq b \iff a \mid b$.

```
   12
  /  \
 4    6
 | / |
 2   3
  \ /
   1
```

Consider a = 4 and b = 6. Since 1 | 4 and 1 | 6, the element 1 is a common lower bound (also 2 | 4, but 2 ∤ 6). So **4 and 6 are compatible** (they share the common lower bound 1) — even though 4 and 6 are **incomparable** (neither divides the other; they sit on different "branches").
Key takeaway: compatible ≠ comparable! Two elements can have a common lower bound without either one being ≤ the other.

Example 2 — a poset with no common lower bound

Consider the poset with c ≤ a, d ≤ b (order drawn upward: 1 on top, a and b in the middle, c and d at the bottom, with c under a only and d under b only). Here a and b have **no common lower bound**, so a and b are **incompatible**.

Definition — Well-ordered set

The linearly ordered set (A, ≤) is a **well-ordered set**, and ≤ a **well-order**, if *every nonempty subset of A has a least element* with respect to ≤.

Note

(ℕ, ≤) is a well-ordered set — this is a foundational assumption (it can't be proven from more basic facts), so we always have at least one well-ordered set to build from.

Theorem

If (A, ≤) is well-ordered and B is a nonempty subset of A, then (B, ≤) is well-ordered.

> *Proof:* Let B ⊆ A, B ≠ ∅. To show B is well-ordered, take any nonempty C ⊆ B. Then C ⊆ A too, and since ≤ well-orders A, C has a least element with respect to ≤. ∎

7. Functions — Formal Definition

Big idea: Forget the "machine that spits out an output" intuition for a second — formally, a function is just a special *set of ordered pairs*: one where every input appears exactly once (paired with exactly one output). The two conditions to check are: (1) every element of A actually gets assigned something, and (2) no element of A gets assigned two different things.

Definition — Function

A **function** is an ordered triple (f, A, B) such that:

1. A and B are sets, and $f \subseteq A \times B$,
2. for every $x \in A$ there is some $y \in B$ such that $(x, y) \in f$, *(every input is used — "for each")*
3. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$. *(each input gets only one output — "uniqueness")*

A is called the **domain** of f, and B its **codomain**. We write (f, A, B) as $\mathbf{f : A \to B}$. If $(x, y) \in f$ we usually write $\mathbf{y = f(x)}$, calling y the **image of x under f**. The set $\{y \in B : \text{there is an } x \in A \text{ such that } y = f(x)\}$ is called the **range** of f.

Notation

- $f : A \to B$ means f is a function from A to B.
- If $a \in A$, we write $b = f(a)$ for the element of B assigned to a.
- We can also write $f : a \mapsto b$, read "f maps a to b."
- If $U \subseteq A$, the **image of U** is $f(U) = \{f(u) \in B : u \in U\} \subseteq B$.
- The image of the whole domain A is exactly the range of f: $f(A) = \text{range}(f) = \text{Im}(f) = \{f(a) : a \in A\}$.

Summary — three ingredients of a function

A function consists of exactly three parts:

1. **Domain** — the input set A,
2. **Codomain** — the "target" set B (where outputs are allowed to live),
3. and a **rule** assigning a unique range object to each object in the domain.

Domain vs. Range vs. Codomain — the classic mix-up: The *codomain* is the set you *declared* outputs would live in (e.g. "$f : \mathbb{R} \to \mathbb{R}$"). The *range* is the set of outputs that *actually get hit*. The range is always a subset of the codomain ($R(f) \subseteq B$), but it doesn't have to equal it — e.g. $f(x) = x^2$ has codomain $\mathbb{R}$ but range only the non-negative reals.

8. Worked Examples — Is it a Function?

Example 1 — yes, it is a function

$f = \{(1, a), (2, b), (3, a)\}$, from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$.
$f \subseteq A \times B$ and $D(f) = A$. Every element of A has a *unique* image in B, so f is a function.
Range: $R(f) = \{f(1), f(2), f(3)\} = \{a, b\}$ (note $R(f) \subseteq B$, and c is simply never hit).

Example 2 — NOT a function (uniqueness fails)

$f = \{(1, a), (2, b), (2, c), (3, a)\}$, proposed from $A = \{1,2,3\}$ to $B = \{a,b,c\}$.
Here $2 \in A$ has **two** images in B (both b and c) — condition 3 (uniqueness) is violated. So f is **not** a function.

Example 3 — NOT a function (domain incomplete)

$f = \{(2, b), (3, a)\}$, proposed from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$.
$f \subseteq A \times B$, but $D(f) = \{2, 3\} \neq A$ — element 1 never gets assigned anything. Condition 2 ("for each") is violated, so f is **not** a function from A to B.

Example 4 — a familiar function

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$ for all $x \in \mathbb{R}$. This is a function, with $D(f) = \mathbb{R}$ and:

$$R(f) = \{f(x) \mid x \in \mathbb{R}\} = \{x^2 \mid x \in \mathbb{R}\} = \{x \mid x \in \mathbb{R} \text{ and } x \geq 0\}.$$

Notice codomain = $\mathbb{R}$ but range = $[0, \infty)$ — the codomain is bigger than the range here, since negative numbers are never actually produced.

9. Graphs of Functions & the Vertical Line Test

Definition — Graph of a function

The **graph** of a function f is the set of all ordered pairs $\{(x, f(x)) \mid x \in D(f)\}$.

Vertical Line Test

A curve in the xy-plane is the graph of a function **if and only if every vertical line intersects it at most once**.

This is just condition 3 (uniqueness) from the definition, shown visually: a vertical line at $x = a$ crosses the curve once for each y-value paired with a. If it crosses twice, that x-value has two different outputs — which breaks the "unique output" rule, so it can't be a function. A line (like $y = x$) passes the test; a circle does **not** (a vertical line through the middle crosses it twice — top and bottom).

10. Equal Functions & Composition

Definition — Equal functions

Two functions $f : A \to B$ and $g : C \to D$ are **equal** if and only if:

1. $A = C$,
2. $B = D$,
3. $f = g$ (meaning $f(x) = g(x)$ for every $x \in A$).

All three conditions matter — two functions with the exact same "formula" but different declared domains/codomains are technically *not* the same function.

Definition — Composition of functions

Let $f : A \to B$ and $g : C \to D$ be functions such that $\operatorname{ran} f \subseteq \operatorname{dom} g$. Then the function $\mathbf{g \circ f : A \to C}$ defined by $(g \circ f)(x) = g(f(x))$ is called the **composition** of f and g.

Note

$D(g \circ f) = \{x \mid x \in D(f) \text{ and } f(x) \in D(g)\}$.

Lemma 1

If f and g are functions such that $\operatorname{ran} f \subseteq \operatorname{dom} g$, then $\mathbf{dom(g \circ f) = dom\ f}$, and $\mathbf{codom(g \circ f) = codom\ g}$.

Lemma 2 — Composition is associative

Let $f : A \to B$, $g : B \to C$, $h : C \to D$ be functions. Then:

$$\mathbf{h \circ (g \circ f) = (h \circ g) \circ f}$$

i.e. the composition of functions is associative — grouping doesn't matter.

Proof: Let $p = h \circ (g \circ f)$ and $q = (h \circ g) \circ f$. To show two functions are equal, we show they act the same way on every input (same domain/codomain follow from Lemma 1). For any x in the shared domain:

$$p(x) = h((g \circ f)(x)) = h(g(f(x)))$$
$$q(x) = (h \circ g)(f(x)) = h(g(f(x)))$$

Since $p(x) = q(x)$ for every x, and the domains/codomains match, $p = q$. ∎

11. Quick-Revision Summary Table

Concept	One-line meaning	Watch out for
Partition	Non-empty subsets that cover A completely with no overlap.	Pieces must be non-empty; check both "covers everything" AND "no overlap."
Equivalence relation ↔ partition	Equivalence classes of ~ always form a partition, and vice versa.	This equivalence goes *both directions* — both are examinable.
Partial order (≼)	Reflexive + antisymmetric + transitive.	Strict < fails reflexivity, so it's *not* itself a partial order.
Poset	A set with a partial order defined on it, written (A, ≼).	Not every pair needs to be comparable — that's the "partial" part.
Comparable	a ≼ b or b ≼ a holds.	Some pairs in a poset can be incomparable.
Weak-Trichotomy	At most one of a ≺ b, b ≺ a, a = b holds.	"At most one," not "exactly one" (unlike ℝ under <).
Least / greatest element	Beats (or loses to) *every* element of A. Unique if it exists.	Doesn't have to exist!
Minimal / maximal element	Nothing strictly below/above it — but can still be incomparable to others.	A poset can have several minimal or maximal elements at once.
Chain	A subset where every pair is comparable.	The whole poset is a chain ⟺ it's linearly ordered.
Compatible	Elements sharing a common lower bound.	Compatible ≠ comparable (see the 4-and-6 divisibility example).
Linear (total) order	Every pair in A is comparable — A itself is one big chain.	(P(A), ⊆) is usually only partial, not linear.
Well-ordered set	Every nonempty subset has a least element.	(ℕ, ≤) is the standard example; subsets of a well-ordered set are also well-ordered.
Function f : A → B	A set of pairs where every x ∈ A gets exactly one y ∈ B.	Two failure modes: missing an input ("for each" fails) or double-assigning one ("uniqueness" fails).
Domain / codomain / range	Domain = inputs; codomain = declared output set; range = outputs actually hit.	Range ⊆ codomain always, but they need not be equal.
Vertical line test	A curve is a function's graph iff every vertical line meets it ≤ once.	Just a visual version of the uniqueness condition.
Equal functions	Same domain, same codomain, same rule on every input.	All three must match, not just the formula.
Composition g∘f	(g∘f)(x) = g(f(x)), needs ran f ⊆ dom g.	Composition is associative: h∘(g∘f) = (h∘g)∘f.

Study tip: When you see a new poset (like a divisibility diagram / Hasse diagram), always ask three questions in order: (1) Is it reflexive/antisymmetric/transitive at all — i.e. is it even a poset? (2) Which pairs are comparable vs. incomparable? (3) Does it have a least/greatest element, and what are its minimal/maximal elements? Walking through these three questions covers almost every exam question on this topic.